

belongs to A and is 0 if a_i does not belong to A .

Example 18: Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, and the ordering of elements of U has the elements in increasing order; that is, $a_i = i$. What bit strings represent the subset of all odd integers in U , the subset of all even integers in U , and the subset of integers not exceeding 5 in U ?

Solution: All odd integers in U : 1010101010 $\{1, 3, 5, 7, 9\}$
 All even integers in U : 0101010101 $\{2, 4, 6, 8, 10\}$
 Subset of integers not exceeding 5 in U : 1111000000

Example 20: The bit strings for the sets $\{1, 2, 3, 4, 5\}$ and $\{1, 3, 5, 7, 9\}$ are 111100000 and 10101010, respectively. Use bit strings to find the union and intersection of these sets.

Solution: Union:
$$\begin{array}{r} 11\ 1110\ 0000 \\ 10\ 1010\ 1010 \\ \hline 11\ 11101010 \end{array} \quad \begin{array}{l} \text{Bitwise OR} \\ \{1, 2, 3, 4, 5, 7, 9\} \end{array}$$

Intersection:
$$\begin{array}{r} 11\ 1110\ 0000 \\ 10\ 1010\ 1010 \\ \hline 10\ 1010\ 0000 \end{array} \quad \begin{array}{l} \text{Bitwise AND} \\ \rightarrow \{1, 3, 5\} \end{array}$$

2.2.5 Multisets

- A **multiset** (short for multiple-membership set) is an unordered collection of elements where an element can occur as a member more than once.
- The notation $\{m_1 \cdot a_1, m_2 \cdot a_2, \dots, m_r \cdot a_r\}$ denotes the multiset with element a_i occurring m_i times, element a_2 occurring m_2 times, and so on. The numbers $m_i, i = 1, 2, \dots, r$ are called the **multiplicities** of the elements $a_i, i = 1, 2, \dots, r$.
- The cardinality of a multiset is defined to be the sum of the multiplicities of its elements.
- The concept dates back to the 12th century work of the Indian mathematician Brahmagupta (12th century). (PvG).
- The **union** of the multisets P and Q is the multiset in which the multiplicity of an element is the maximum of its multiplicities in P and Q .